Contents

A local, generally covariant field theory of modified inertia

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The theory in one box

$$\boxed{S \;=\; S_{\rm EH}[g] \;+\; S_{\rm kh}[g,T] \;+\; S_\chi[g,u,\chi] \;+\; S_{\rm m}[g,T,\chi,x]}$$

$$S_{\rm kh} = \frac{1}{16\pi G}\int N\sqrt{h}\Big[K_{ij}K^{ij} - \lambda K^2 + \xi R^{(3)} + \eta\, a_i a^i\Big], \qquad a_i = \partial_i \ln N, \qquad n_\mu = -\frac{\partial_\mu T}{\sqrt{-(\partial T)^2}}$$

$$(u^\mu\partial_\mu + m)^2\chi = \frac{g\,|a|}{c}, \qquad S_{\rm m} = -mc^2\int\Big[\mu(\chi)\,d\tau + (1-\mu(\chi))\,dt\Big]$$

Three things are true of this action that were not true of the worldline construction it completes [1]:

1. **It is local.** The memory kernel is gone — replaced by an auxiliary field χ whose retarded Green's function *is* that kernel (§2).
2. **It is generally covariant.** The preferred frame is the normalised gradient of a scalar, so the Lorentz violation is spontaneous rather than stipulated (§3).
3. **Its propagating content is 2 + 1 + 0 = 3 modes, and all three are healthy** in an explicit nonempty window (§4), **with a strong-coupling scale that clears every scale the theory is applied at by dozens of orders, and no Vainshtein-type screening anywhere** (§5).

And the acceleration scale is no longer the first moment of a postulated kernel:

$$\boxed{a_0 = \tfrac{2}{3}\, c\, \frac{m^2}{g}}$$

the ratio of the auxiliary field's mass squared to its coupling. Together with the companion no-go result [2] — that this moment is logarithmically divergent, so it cannot be computed from a free-field correlator — this means g/m^2 is a *renormalised coupling*. Which is what every coupling in every local field theory is.

The honest caption, which must travel with the box. a_0's *value* is still not derived, and the covariantisation *added* two free parameters (λ, η) rather than removing any. The theory is complete in the sense that a local, covariant, ghost-free action exists, has a strong-coupling scale far above every regime it is applied in, and yields MOND; it is not complete in the sense of predicting its own constants. And it says nothing whatever about particle physics — see §9.

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1. What was missing

Reference [1] gives a worldline action reproducing Milgrom's modified-inertia relation via the *rapidity gap*, $\cosh\theta = -u{\cdot}u'/c^2$, with the memory encoded in a kernel $K(s)$ whose first moment satisfies $M_1 = \frac{2}{3}c/a_0$. Three things were owed:

- **Nonlocality.** $\Theta(\tau) = \int_0^\infty ds\, K(s)\, \theta(\tau, \tau - s)$ has no local derivative expansion, so there is no effective-field-theory power counting and no Hamiltonian.
- **Ghosts.** The equation of motion is fourth order in x. Ostrogradsky's theorem does not apply, because it requires a *local* higher-derivative Lagrangian — but that is a statement about the theorem's silence, not a proof of ghost-freedom.
- **A stipulated frame.** The action uses dt, the cosmological time, and a fixed timelike n^μ. That breaks general covariance by hand, and it makes the paper's Lorentz-violation prediction dismissible as an artefact of choosing n.

This paper discharges all three. Each section's claims are produced by a committed script that exits non-zero on any internal failure and carries negative controls that must trip (§10).

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2. Step 1: the kernel is a local field

The reduction. For a general worldline the rapidity gap obeys $\theta(\tau, \tau - s) = (s/c)\,|a(\tau - s/2)| + O(s^3)$ [1] — the gap across an interval is s/c times the acceleration magnitude at the **midpoint**. So Θ carries both a factor of s and a lag of $s/2$. Substituting $s = 2u$ collapses *both* into one kernel:

$$\Theta(\tau) = \int_0^\infty du\, G(u)\, J(\tau - u), \qquad G(u) = 4u\, K(2u), \qquad J = |a|/c.$$

For the minimal causal kernel $K(s) = (N/\lambda)e^{-s/\lambda}$ this is

$$G(u) = g\,u\,e^{-mu}, \qquad m = \frac{2}{\lambda}, \qquad g = \frac{4N}{\lambda},$$

whose zero-frequency gain is $\int_0^\infty G\,du = g/m^2 = N\lambda = M_1$ **exactly**: the kernel's first moment *is* the local operator's DC gain.

Theorem 1. *$g\,u\,e^{-mu}$ is the retarded Green's function of $(d/d\tau + m)^2$.*

Verified three independent ways: it solves the homogeneous equation for $u > 0$; its jump conditions are $G(0) = 0$, $G'(0) = g$, so $(d/du + m)^2 G = g\,\delta(u)$; and its Laplace transform is $g/(p+m)^2$. Hence

$$\boxed{\ddot{\chi} + 2m\dot{\chi} + m^2\chi = \frac{g\,|a|}{c}, \qquad \Theta = \chi}$$

— **a damped oscillator whose friction term *is* the memory.** This is closed numerically as well as formally: integrating the local ODE forward with retarded data reproduces the nonlocal convolution to 10^{-23}, and perturbing m by 5% breaks that agreement by 5×10^{-2}.

The damping is critical, exactly, and it is forced. The discriminant of $p^2 + 2mp + m^2$ vanishes identically; there is a double root at $p = -m$. And for $K \sim s^k e^{-s/\lambda}$ the local operator has order $k + 2$, so the exponential kernel gives the *minimal* localisation. The extra factor of u traces to the rapidity gap being **linear in** s — which is the same fact that, in the companion paper [2], places the required kernel moment exactly on the pole of ζ. One structural feature, three consequences.

The ghost question becomes a computation. Enforcing the constraint with a multiplier π, the adjoint of $(d/d\tau + m)^2$ is $(d/d\tau - m)^2$ — *anti*-damped, the Bateman mirror — and the (Θ, π) kinetic form is off-diagonal, with eigenvalues $\pm\frac{1}{2}$ and signature $(+, -)$. By the naive criterion, a ghost. **It is not a new degree of freedom.** π is a *costate*: it carries a final-value condition rather than Cauchy data, it decays going backward ($|\pi| \sim e^{-m(T-\tau)}$, bounded by 1 on the physical interval), and the count closes exactly — 2 conditions on Θ plus 2 on $\pi = 4$, which is the order of the fourth-order equation of motion in x already derived in [1]. The indefinite metric is the Keldysh $(-)$ branch of the in-in formalism the construction already uses. This *replaces* the "Ostrogradsky is silent" argument with a positive statement.

a_0 **becomes a coupling ratio.** From $a_0 = \frac{2}{3}c/M_1$ and $M_1 = g/m^2$:

footing	M_1 (Gyr)	λ (yr)	m (s^{-1})	g (s^{-1})	N	a_0 rebuilt
canonical	67.65	3.2×10^4	1.9805×10^{-12}	8.374×10^{-6}	2.114×10^6	9.3619×10^{-11}
ALT ($\times 1.2048$)	56.15	3.2×10^4	1.9805×10^{-12}	6.951×10^{-6}	1.755×10^6	1.1279×10^{-10}

reproducing a_0 to 10^{-25} on both footings, with the implied kernel weight matching the independently derived bound $N \geq 2.1 \times 10^6$ of [1].

Costs, in the same breath. (i) The localisation is exact for the *midpoint* form and only third-order accurate for the exact bilocal action; galactically $\lambda\Omega = 8.8 \times 10^{-4}$, so the relative error is $O((\lambda\Omega)^2) = 7.7 \times 10^{-7}$ — small, not zero. (ii) In x the localised system is *second*-derivative, because the source is $|a|$, so **Ostrogradsky is not evaded in the localised writing.** The exact action contains only u, so this is an artefact of the midpoint expansion — which is precisely why "Ostrogradsky-free" and "fourth-order equation of motion" coexisted in [1] rather than contradicting each other. (iii) **Exact localisation of the full bilocal** $\cosh^{-1}(-u\cdot u'/c^2)$ **is not available by this route at all**: the auxiliary-propagator trick requires linearity in the delayed field. A genuine limitation.

Covariantly, $(u^\mu\partial_\mu + m)^2\Theta = g|a|/c$ is a scalar advected by the matter flow obeying second-order relaxation — structurally the Israel–Stewart class [3], whose purpose was to restore the causality and hyperbolicity that first-order dissipative hydrodynamics destroys. A route and an analogy; no Israel–Stewart result is imported.

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3. Step 2: the preferred frame becomes dynamical

Replace n^μ by the normalised gradient of a scalar khronon, $n_\mu = -\partial_\mu T/\sqrt{-(\partial T)^2}$. It is unit timelike by construction, and invariant under the reparametrisation $T \to f(T)$: carrying $f'(T)$ as a positive symbol, the derivative of n with respect to it vanishes identically. So the **foliation is physical and the labelling is gauge**.

Theorem 2 (the vorticity vanishes identically). *For any n built from a gradient, $\partial_{[\mu}n_{\nu]} = A_{[\mu}n_{\nu]}$ with $A = -\partial \ln N$, and the spatial projector annihilates n; hence $\omega_{\mu\nu} = 0$ in every metric.*

Verified termwise on a generic T and generic N — T's second derivatives cancel — and the Christoffels drop out of the antisymmetrisation, so the result is metric-independent. **Consequence: the realisation is the hypersurface-orthogonal (Hořava) sub-case of Einstein-aether [4,5,6], not the general aether, and the spin-1 sector is removed entirely.** A prespecified non-gradient decoy field has nonzero curl, confirming the theorem is about gradients and not about the antisymmetrisation.

Theorem 3 (the khronon's acceleration is the Newtonian field).

$$a^\mu[n] = \partial^\mu\Phi \qquad \text{(linear order, static weak field)}$$

Computed from the Christoffels of an explicit metric, not quoted; it is purely spatial ($a^t = 0$) and equals the log-lapse gradient, as the general $K/\omega/a$ decomposition requires. Three prespecified decoys — $2\partial\Phi$, $-\partial\Phi$, $\partial\Phi/2$ — are all rejected, so this measures the coefficient *and* the sign.

This matters more than the repair does. Before covariantisation, the only acceleration in the theory was the particle's own $|a|$. MOND is a relation between g_{obs} and g_{bar}, and g_{bar} now exists as a *geometric object*. §6 draws the consequence.

Nothing is lost. In the gauge $\sqrt{-(\partial T)^2} = 1$ with $T = t$, the covariant coupling $\sqrt{(u\cdot n)^2}$ equals γ **exactly** — so the action of [1] is the *unitary gauge* of the khronon action. The covariantisation is a completion, not a modification. It remains CPT-even for the same reason Form III of [1] was: $(u\cdot n)^2$ is invariant under $n \to -n$ while $(u\cdot n)$ flips.

The couplings collapse from four to three. With $\omega = 0$ the decomposition $\nabla_\mu n_\nu = K_{\mu\nu} - n_\mu a_\nu$ gives $(\nabla_\mu n_\nu)(\nabla^\mu n^\nu) = K\cdot K - a\cdot a$ and $(\nabla_\mu n_\nu)(\nabla^\nu n^\mu) = K\cdot K$, so c_1 and c_3 differ only by the acceleration term, which is degenerate with c_4:

$$\sum_i c_i T_i = (c_1{+}c_3)\, K\cdot K + (c_4{-}c_1)\, a\cdot a + c_2\,(\mathrm{tr}\,K)^2$$

with Jacobian rank 3 — matching the three parameters of the infrared limit of non-projectable Hořava gravity [4]. Reinstating a vorticity term makes T_1 and T_3 independent again with coefficient $(c_1 - c_3)$, confirming the collapse is a consequence of Theorem 2 and not an algebraic accident. **Propagating content: 2 tensor + 1 scalar, no spin-1.**

And the Lorentz-violation prediction becomes a real coupling. $|B| = (1-\mu)/2 \approx a_0^2/8g^2$ [1] previously multiplied a stipulated n; with n dynamical it is the coefficient of a matter–khronon coupling of CPT-even $c^{\mu\nu} \sim B\, n^\mu n^\nu$ type, so the g^{-2} scaling can no longer be dismissed as an artefact of the choice of frame.

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4. The spin-0 health check

This is the calculation that could have killed §3, so it is done from scratch rather than by quoting a dispersion relation. Perturbing in the scalar sector, $N = 1+\alpha$, $N_i = \partial_i B$, $h_{ij} = (1+2\zeta)\delta_{ij}$, and working in Fourier so every integration by parts is algebra:

$$S_2 = \int dt \Big[3(1{-}3\lambda)\dot\zeta^2 + 2(1{-}3\lambda)k^2\dot\zeta B + (1{-}\lambda)k^4 B^2 + 2\xi k^2\zeta^2 + 4\xi k^2\alpha\zeta + \eta k^2\alpha^2 \Big]$$

with α and B carrying no time derivative — they are constraints.

The GR check passes first. The α constraint gives $\alpha = -2\xi\zeta/\eta$, and **at $\eta = 0$ it degenerates to $4\xi k^2\zeta = 0$, forcing $\zeta = 0$: general relativity has no propagating scalar.** Reproduced, not assumed — so the mode found below is genuinely the Lorentz-violating one. Eliminating both constraints leaves $A = 2(1{-}3\lambda)/(1{-}\lambda)$ and

$$\boxed{c_s^2 = \frac{\xi\,(2\xi-\eta)(1-\lambda)}{\eta\,(1-3\lambda)} \overset{\xi=1}{=} \frac{(2-\eta)(\lambda-1)}{\eta\,(3\lambda-1)}}$$

The tensor sector independently gives $\frac{1}{4}\dot\gamma^2 - \frac{\xi}{4}k^2\gamma^2$: graviton speed$^2 = \xi$ with a *positive* kinetic term, which is what anchors the overall sign convention — and means **GW170817 makes $\xi = 1$ a measurement to $\sim 10^{-15}$, not a choice.**

The window, and the two diseases are one band:

- **No ghost** $\iff A > 0 \iff \lambda > 1$ or $\lambda < 1/3$.
- **No gradient instability** $\iff c_s^2 > 0 \iff 0 < \eta < 2$.

Both exclude **exactly** $1/3 < \lambda < 1$, where the mode is simultaneously a ghost and gradient-unstable. The healthy region is nonempty and explicit:

(λ,η)	$(2,1)$	$(2,\frac{1}{2})$	$(1.0001, 10^{-4})$	$(0.2, 1)$	$(0.2,\frac{1}{2})$
c_s^2	0.20	0.60	0.9998	2.0	6.0

Four prespecified sick decoys are rejected; rescaling ζ leaves c_s^2 unchanged, so it is a physical speed; and flipping the action's overall sign makes the *graviton* go bad, confirming the no-ghost criterion is anchored to the tensor sector.

And observation points the right way. Preferred-frame PPN pushes both $(\lambda - 1)$ and η small, and in that corner

$$c_s^2 \;\longrightarrow\; \frac{\lambda - 1}{\eta} \qquad \text{exactly}$$

— **only the ratio survives**, so smallness alone drives c_s neither to zero nor to infinity. The vacuum Cherenkov bound [7] requires the scalar to be at least luminal, since a subluminal mode would be radiated by ultra-high-energy cosmic rays. So it reduces to a single inequality,

$$\lambda - 1 \;\geq\; \eta \;>\; 0,$$

and superluminal propagation is *safe* here precisely because there is a preferred frame. **PPN and Cherenkov make independent demands: no conflict.**

Against interest. (i) **The health is achieved by choice, not predicted.** §3 bought general covariance at the price of two new free parameters, and nothing in modified inertia forces either; "the scalar is healthy" means "a viable region exists", which is much weaker than a prediction. The theory *gained* parameters. (ii) **The $\eta \to 0$ corner that PPN prefers is the known strong-coupling corner** ($c_s^2 \to \infty$ at fixed λ): the strong-coupling scale of the infrared non-projectable theory falls as the Lorentz-violating couplings do [5], so PPN safety pushes the cutoff *down*. **That scale is not computed here**, and it is the sharpest remaining worry about the covariantisation. (iii) The α_1, α_2 formulas are not derived; only their qualitative pressure is used, and the 10^{-7} figures are standard solar-system orders [8].

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5. Strong coupling and the static nonlinearity

5.1 The strong-coupling scale

Section 4 leaves one danger: preferred-frame PPN pushes $(\lambda - 1)$ and η small, and a kinetic term proportional to a small number is exactly when self-interactions turn on early. That was the sharpest open risk in v1 of this paper, and it is now computed.

Why the danger is real, stated precisely. Restore the khronon fluctuation, $T = t + \pi$. Then

$$\ln N = -\dot{\pi} + \tfrac{1}{2}\dot{\pi}^2 + \tfrac{1}{2}(\partial\pi)^2 + O(\pi^3), \qquad a_i = -\partial_i\dot{\pi} + O(\pi^2), \qquad K_{ij} = -\partial_i\partial_j\pi + O(\pi^2),$$

and **at $\lambda = \xi = 1$, $\eta = 0$ the khronon action vanishes identically** — the K sector gives $K{\cdot}K - K^2 = k^4\pi^2 - k^4\pi^2 = 0$, so π is pure gauge in general relativity, as it must be. A prespecified $\lambda = 3/2$ decoy does *not* cancel, so this detects the GR limit rather than an artefact of the substitution. Everything that survives is therefore proportional to the small parameters:

$$S_2 \sim M_{\rm Pl}^2\Big[-\delta\,(\partial^2\pi)^2 + \eta\,(\partial_i\dot{\pi})^2 \Big], \qquad \delta = \lambda - 1.$$

And an independent cross-check falls out. That action's dispersion is $\eta\,\omega^2 k^2 = \delta\, k^4$, i.e.

$$c_s^2 = \frac{\delta}{\eta} = \frac{\lambda - 1}{\eta}$$

which is **exactly** the PPN-corner limit of §4 — obtained there from the unitary-gauge ζ with the ADM constraints eliminated, and here from the Stückelberg π in flat space. Two independent gauges and variables, one answer; a prespecified decoy $\delta/(2\eta)$ is rejected.

The static nonlinearity of the η sector vanishes. Canonical normalisation from the η term is $\pi_c = \sqrt{\eta}\, M_{\rm Pl} k\, \pi$, and the leading self-interaction is

$$-2\dot\pi(\partial_i\dot\pi)^2 - 2\,\partial_i\dot\pi\,\partial_j\pi\,\partial_i\partial_j\pi$$

— **every term carries a time derivative, so the whole cubic part vanishes for static configurations.** There is no static Vainshtein-type screening radius from this sector; the nonlinearity needs time dependence to switch on. A decoy cubic $(\partial_x\pi)^3$ does not vanish statically, so this is a property of the actual terms. Derivative counting on the same terms gives cubic/quadratic $= c_s E/(\sqrt{\eta} M_{\rm Pl})$, hence

$$\boxed{\Lambda_{\rm sc} \sim \frac{\sqrt{\eta}\, M_{\rm Pl}}{c_s} \;\approx\; 7.7 \times 10^{14}\ \text{GeV} \quad (\eta = 10^{-7})}$$

So the PPN-preferred corner does lower the cutoff, but only as $\eta^{1/2}$, and from the Planck mass.

The comparison that settles it, built to be robust to the power. The theory is applied at absurdly low energies. Scanning $\Lambda_{\rm sc} = \eta^p M_{\rm Pl}$ over $p = \frac{1}{2}, 1, 2, 3, 4$ — deliberately allowing powers far worse than the one derived:

scale where the theory is applied	energy	$\Lambda_{\rm sc}/E$ at $p = \frac{1}{2}$	at $p = 4$
Milky Way orbital frequency	5.7×10^{-31} eV	1.4×10^{54}	4.3×10^{29}
Milky Way inverse size (8.2 kpc)	7.8×10^{-28} eV	9.9×10^{50}	3.1×10^{26}
solar system, 1/AU	1.3×10^{-18} eV	5.8×10^{41}	1.9×10^{17}
laboratory, 1/m	2.0×10^{-7} eV	3.9×10^{30}	1.2×10^{6}

$\Lambda_{\rm sc}$ **exceeds every scale at which the theory is applied, for every power tested** — 26 orders of galactic margin even at the pessimistic $p = 4$, and 50 at the derived $p = \frac{1}{2}$. So the strong-coupling scale bears on whether this is a UV-complete *quantum* theory, which it never claimed to be, and **not** on the phenomenology.

Against interest, and this corrects a claim of my own. A first draft of the accompanying script asserted that the cutoff clears every scale *including* the LHC (1.4×10^{13} eV). **That is false: at $p \geq 3$ the cutoff falls below collider energies**, so the khronon effective theory would not cover the LHC at those powers. It is harmless only because the matter–khronon coupling there is $|B| \sim a_0^2/8g^2 = 1.1 \times 10^{-23}$, and the *derived* power $p = \frac{1}{2}$ clears the LHC by 10 orders — but the claim needed correcting rather than softening. Two further caveats: **only the scaling is computed, not the coefficient** (§5's table is built to be insensitive to it, but a factor of 100 in the prefactor is excluded by nothing here); and **the δ-sector's static nonlinearity is not analysed** — the η sector's static cubic dies, but the $(\partial^2\pi)^2$ sector's cubics need not, and a Vainshtein radius from *that* sector is uncomputed. That is now the honest residual. The analysis is flat-space throughout.

A note on which theory this is. The notorious $\lambda \to 1$ strong coupling belongs to *projectable* Hořava gravity. The non-projectable "healthy extension" carrying the $a_i a^i$ term is the known repair [5] — and §3 landed on it **by theorem** rather than by choice, since the vorticity of a gradient-built n vanishes identically.

5.2 The static nonlinearity of both sectors

§5.1 established that the η sector's cubic self-interaction vanishes for static configurations, and flagged the δ sector — the $-\delta K^2$ term — as unanalysed. That was v2's sharpest remaining gap. It closes by a parity theorem.

Theorem 4 (K is odd in π, to all orders). *For a static khronon $T = t + \pi(\mathbf{x})$, the unit normal is $n^\mu = (1/w, -\partial_i\pi/w)$ with $w = \sqrt{1-(\partial\pi)^2}$. Under $\pi \to -\pi$ the normalisation w is invariant — it depends on $(\partial\pi)^2$ — while $\partial_i\pi$ flips. Hence $K[-\pi] = -K[\pi]$ exactly.*

Verified on the full three-dimensional closed form with **no series truncation**. In one dimension the divergence has an exact closed form that makes the parity manifest:

$$K = -\frac{\pi''}{(1-\pi'^2)^{3/2}}$$

— odd numerator, even bracket. Its expansion has **zero even orders** (checked through fifth): $-\pi''$, then $-\frac{3}{2}\pi''\pi'^2$, then quintic. And the general three-dimensional cubic $-\frac{1}{2}\partial^2\pi(\partial\pi)^2 - \partial_i\pi\,\partial_j\pi\,\partial_i\partial_j\pi$ reduces in one dimension to exactly $-\frac{3}{2}\pi''\pi'^2$, so the 3-D formula is *validated against* the exact 1-D result rather than asserted. Inserting a genuine even piece into n^i by hand gives K a nonzero quadratic term, so the theorem is a measurement and not an algebraic accident.

Corollary. K^2 is even, so the δ sector has no static cubic; its leading static self-interaction is quartic, with

$$\frac{\text{quartic}}{\text{quadratic}} = 3\pi'^2 \qquad \text{(one dimension)}$$

i.e. of *order* $(\partial\pi)^2$ with an $O(1)$ coefficient. (A first draft of the accompanying script wrote "exactly $(\partial\pi)^2$"; the coefficient is 3, and the script records the correction.)

And on the aligned static foliation it vanishes outright. For a static, shift-free metric the constant-t surfaces have $K_{ij} = \frac{1}{2N}(\dot h_{ij} - D_iN_j - D_jN_i) = 0$ identically, so both $K{\cdot}K$ and K^2 vanish at *every* order — only $\eta\, a_i a^i$ survives. And $\delta(K^2)/\delta T = 2K\,\delta K$ vanishes with K, which makes that foliation a **solution** of the K sector rather than an imposed ansatz. A time-dependent metric gives $K_{ij} \neq 0$, so this is a property of staticity and not of the formula.

The nonlinearity, priced. $|\partial\pi|$ is the tilt of the khronon foliation relative to the local frame, i.e. $\sim v/c$:

setting	v	$(\partial\pi)^2$
solar system vs the CMB frame	369.8 km/s	1.5×10^{-6}
galactic rotation	220 km/s	5.4×10^{-7}
cluster velocities	1000 km/s	1.1×10^{-5}
a relativistic probe	$0.1c$	1.0×10^{-2}

So there is no Vainshtein-type screening radius: the nonlinearity never reaches $O(1)$. It would require $|\partial\pi| \to 1$ — a foliation boosted at near-light speed relative to the local frame — which happens near a black-hole horizon and nowhere else.

Combining the two sectors. The η cubic dies by carrying a time derivative; the δ cubic dies by parity in π. Two different mechanisms, so **the leading static self-interaction of the whole khronon sector is quartic** — worth a factor of 811 over a surviving cubic (1.5×10^{-6} instead of 1.2×10^{-3}). The result is not cosmetic.

What remains, and it is narrower than what it replaces. (i) **The full T field equation around a real source is not solved.** The aligned foliation is shown consistent with the K sector and the nonlinearity is priced *given* $|\partial\pi| \sim v/c$, but $\pi(r)$ is not obtained, so a larger $|\partial\pi|$ than the kinematic estimate is not excluded. (ii) $|\partial\pi| \to 1$ **near a black-hole horizon is exactly where this analysis fails**; universal horizons in Lorentz-violating gravity [12] are a real known issue and are not addressed. (iii) The η sector's quartic is not computed either — only that its cubic vanishes.

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6. Step 3: MOND from the joint field equations

The metric sector stays Newtonian — and the constraint that made the khronon healthy is what guarantees it. Modified inertia *presupposes* an unmodified gravitational field: its entire content is a modified *response* to a Newtonian Φ. Khronon corrections to Φ are $O(\eta, \lambda - 1) \lesssim 10^{-7}$, while MOND phenomenology needs Φ_{bar} only to $\sim 1\%$. **Five orders of margin.** So §4's constraint is not a cost — it is a consistency requirement the framework needed anyway. A decoy $\eta = 0.5$ would leave the metric wrong by 50%, so this is a real comparison.

The mass question, confronted. From $E = mc^2[1 + \mu(\gamma - 1)]$ [1]: the rest energy is mc^2, independent of μ, while the kinetic term is $m\mu v^2/2$. Therefore

$$m_{\mathrm{grav}} = m, \qquad m_{\mathrm{inert}} = m\mu, \qquad \text{and they differ.}$$

That is not a defect to be explained away — **it is the definitional content of modified inertia.** And the weak equivalence principle survives: $m\mu a = m g_{\mathrm{bar}}$ gives $\mu a = g_{\mathrm{bar}}$ with no reference to mass or composition, so universality of free fall is exact. What is violated is the *equality* $m_i = m_g$. A decoy μ depending on the mass leaves m in the answer, so the cancellation is a real property of $\mu(\Theta)$. The consequence runs in the framework's favour: a galaxy's **gravitating** mass is its **baryonic** mass. Energy-momentum conservation for a time-nonlocal modified-inertia theory is cited [9], not proved here.

The chain closes. With the Newtonian metric established independently, the worldline equation $d(m\mu v)/dt = -m\nabla\Phi$ gives

$$\mu(g_{\rm obs}/a_0)\, g_{\rm obs} = g_{\rm bar} \implies g_{\rm obs} = \sqrt{a_0 g_{\rm bar}} \implies v^4 = GMa_0$$

exactly — flat rotation curves and the baryonic Tully–Fisher relation — with **no new parameter**, since $a_0 = \frac{2}{3}c\,m^2/g$ is inherited from §2. A decoy deep limit $\mu \sim Y^2$ fails to reproduce $v^4 = GMa_0$, so this tests the interpolation's deep power and not the algebra of substitution. The closure is not circular: the Newtonian metric came from §5's first paragraph, not from the MOND relation.

The BTFR normalisation, both ways. The framework predicts $A_{\rm fw} = 1/(Ga_0) = 80.5\,M_\odot(\text{km/s})^{-4}$ against McGaugh's fitted $A = 47$ [10] — a ratio of 1.712, i.e. **0.234 dex in mass**, or $v_c = 105.6$ km/s at $10^{10}M_\odot$ versus 120.8. Three things must be said together:

- **Against interest: the ALT footing fits better** (ratio 1.421 vs 1.712).
- **But it is not a test of a_0.** $M_{\rm bar}$ scales with the stellar mass-to-light ratio, and $A = 47$ was obtained at $\Upsilon_{3.6} = 0.5$; the framework needs $\Upsilon_{3.6} = 0.856$ — high, yet inside the literature spread and close to the 0.70 that an independent RAR fit of this corpus prefers. The intercept is Υ-degenerate.
- **And the Υ-free estimator accepts the canonical value.** The gas-dominated a_0-line box $[0.84, 1.36] \times 10^{-10}$ contains 9.3619×10^{-11}.

So the offset is a mass-to-light question, not an a_0 question: **neither a win nor a deficit.**

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7. A fork the covariantisation exposes

This is the genuinely new physical content, and it is a consequence of Theorem 3. The covariant theory possesses **two** acceleration scalars: the particle's own $|a|$, and the khronon's $|a[n]| = g_{\rm bar}$. So Θ may be sourced by either:

source of Θ	theory
$\Theta[\,\lvert a\rvert\,]$	**pure modified inertia** — the construction of [1]
$\Theta[\,\lvert a[n]\rvert\,]$	**a third theory** — external-field-driven; neither pure MI nor AQUAL

The second branch shares the same interpolation function and the same a_0, and is a genuinely different theory. **The worldline formulation could not even express this fork**, because it had no covariant object equal to $g_{\rm bar}$.

They are observationally distinguishable. In a directional external-field test — the aligned versus anti-aligned asymmetry of rotation curves relative to the external field direction — **pure modified inertia predicts exactly zero**, while an external-field-driven Θ does not. The fork is exposed here and **not resolved here**.

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8. Limitations

- **Single-stream: a formulation preference, not a restriction on the physics.** A first draft of this paper called this the sharpest gap. That was wrong, and the correction is recorded rather than absorbed. $\Theta(\tau)$ is an integral over *the particle's own past*, so $\ddot{\chi} + 2m\dot{\chi} + m^2\chi = g|a|/c$ is an ODE in the particle's own proper time: χ is a **per-worldline internal variable**, like a spin or an internal clock, and two stars crossing at a point simply carry different χ. Multi-stream is then a non-issue. And the metric sector never sees χ at all, because $m_{\rm grav} = m$ is μ-independent (§6) — the first χ-dependent source term is the kinetic $m\mu v^2/2$, suppressed by $(v/c)^2 = 5.4 \times 10^{-7}$ at galactic speeds. So the single-stream restriction binds only a *continuum rewriting of* χ *undertaken for its own sake*: it does not restrict test-particle dynamics, which is what rotation curves are, and it does not touch the field equations.
- **Strong coupling and the static nonlinearity are now both computed** (§5) and neither threatens the phenomenology. Three narrower residuals remain in their place: only the *scaling* of $\Lambda_{\rm sc}$ is derived and not its coefficient; **the full T field equation around a real source is not solved**, so a larger $|\partial\pi|$ than the kinematic v/c estimate is not excluded; and the η sector's quartic is uncomputed.
- **Strong field and near-horizon behaviour is the one regime where the analysis genuinely fails.** $|\partial\pi| \to 1$ invalidates §5.2's expansion, and **universal horizons** in Lorentz-violating gravity [12] are not addressed. This is now the sharpest structural gap.
- **Nothing new is derived.** a_0 is the coupling ratio of §2 and μ's shape is the $\alpha = 2$ interpolation that solar-system ephemerides force [1]. Steps 1–3 establish *consistency and locality*, not predictive content.
- **The full bilocal cannot be localised** by the auxiliary-field route (§2).
- Strong fields, black-hole universal horizons, and nonlinear stability: not addressed. The analysis of §4 is flat-space, quadratic order, scalar sector.
- **Ostrogradsky is not evaded in the localised writing** (§2), only in the exact one.

What a referee should attack first

1. **Whether Θ should be sourced by $|a|$ at all** — §6's fork is a genuine ambiguity of the covariant theory, not a detail.
2. **The near-horizon regime** — $|\partial\pi| \to 1$ breaks §5.2, and universal horizons are unaddressed.
3. **The two new parameters λ, η** — whether a theory that gained parameters to become covariant has really gained ground.

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9. What is not claimed

- **a_0's value is not derived**, and $\kappa = \frac{1}{2}$ in $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ remains **fitted**. The companion no-go [2] shows *why* a free-field correlator cannot supply it.
- **The theory acquired two parameters, not fewer.** General covariance and ghost-freedom were bought with λ and η, whose only constraints are the health window and PPN.
- **No claim is made regarding particle physics, the Standard Model, or unification.** The author publicly withdrew earlier statements of that kind and does not restate them. A complete field theory of modified inertia is not a theory of everything, and nothing here is evidence for one.
- **The interpolating function is not new.** $\nu = \sqrt{1 + 1/y}$ is Eq. (9) of Milgrom (1999).

- The khronon sector is standard technology [4,5,6]; what is new here is its use to complete *this* construction, Theorem 3, and the fork of §6.

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10. Reproducibility, and version history

Every quantitative claim above is produced by a committed script that exits non-zero if any internal consistency check fails, and each carries negative controls that must trip. All four are included:

- `mi_kernel_localisation_2026.py` — §2, 27/27 checks.
- `mi_khronon_covariantisation_2026.py` — §3, 26/26 checks.
- `mi_khronon_spin0_health_2026.py` — §4, 30/30 checks.
- `mi_khronon_strong_coupling_scale_2026.py` — §5.1, 24/24 checks. **New in v2.**
- `mi_khronon_delta_sector_static_2026.py` — §5.2, 22/22 checks. **New in v3.**
- `mi_step3_joint_field_equations_2026.py` — §§6–8, 32/32 checks.

Changes in v3

1. **New §5.2: the static nonlinearity of both sectors**, which was v2's sharpest named gap. It closes by a **parity theorem** — K is odd in π to all orders, so K^2 is even and the δ sector has **no static cubic**. Its leading static self-interaction is quartic.
2. On the aligned static foliation the K sector vanishes at every order, and it is a *solution* rather than an ansatz, since $\delta(K^2)/\delta T \propto K$.
3. The nonlinearity is priced at $(\partial\pi)^2 \sim (v/c)^2 \leq 1.1 \times 10^{-5}$ everywhere the theory is applied: **no Vainshtein-type screening radius exists.**
4. Combining with §5.1, **the leading static self-interaction of the whole khronon sector is quartic** — the two cubics vanish for two different reasons.
5. **A claim of the author's own is corrected:** a draft wrote the quartic/quadratic ratio as "exactly $(\partial\pi)^2$"; the coefficient is 3.
6. The limitations list is updated again: the δ-sector item is discharged, and the **near-horizon regime with universal horizons** becomes the sharpest structural gap.
7. Nothing else changes. a_0's value is still not derived and no claim is made about particle physics.

Changes in v2

1. **New §5: the strong-coupling scale**, which was v1's sharpest named risk. It is computed and **does not threaten the phenomenology**: $\Lambda_{\mathrm{sc}} \sim \sqrt{\eta}M_{\mathrm{Pl}}/c_s$, and the conclusion survives a scan over powers η^p, $p = \frac{1}{2} \dots 4$.
2. **A cross-check gained, not assumed:** the Stückelberg π formulation reproduces $c_s^2 = (\lambda - 1)/\eta$, which §4 obtained from unitary-gauge ζ — two gauges, one answer.
3. **The η-sector static nonlinearity is shown to vanish**, so there is no static Vainshtein-type screening from it.
4. **A claim of the author's own is corrected rather than softened:** a first draft asserted the cutoff clears *every* scale including the LHC. It does not — at $p \geq 3$ it falls below collider energies. Stated in §5.
5. The limitations list is updated: the strong-coupling item is discharged and replaced by the narrower and now-sharpest gap, **the δ-sector's static nonlinearity**.

6. Nothing else changes. Every equation, result and caveat of v1 stands, including that a_0's value is not derived and that no claim is made about particle physics.

Both a_0 footings (canonical $\rho_{\rm DE}$ with cH_Λ; ALT $\times 1.2048$) are carried on every dimensionful number.

AI-assistance disclosure. Portions of the analysis, numerical verification and drafting were carried out with the assistance of a large language model (Anthropic Claude). The author directed the work, specified and reviewed every load-bearing calculation, and takes full responsibility including for any errors. No AI system satisfies the criteria for authorship and none is listed as an author. Several intermediate claims produced during the work were found to be incorrect and were withdrawn before this deposit — among them a past-directed sign for n_μ, an assumed rather than derived reparametrisation invariance, an asserted BTFR band that was replaced by the computed value, the overstated single-stream limitation withdrawn in §8, an assertion that the strong-coupling scale clears collider energies at every power (corrected in §5.1), a cubic-order counter that returned zero terms and thereby made a check pass vacuously (found and replaced), and a quartic/quadratic ratio written as exact when its coefficient is 3 (corrected in §5.2).

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